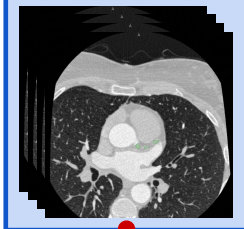


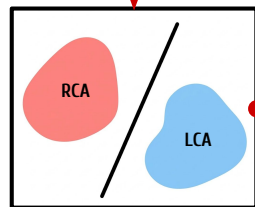
Input Data

SEGMENTATIONS



Load Binary Mask

Split RCA/LCA
Using Mass Center



For Each Component

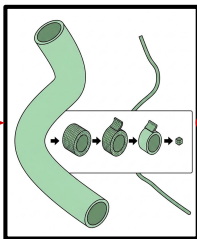
Mesh Creation
& Smoothing

```
vmtkMarchingCubes()  
vmtkSurfaceSmoothing()
```

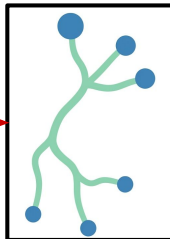
3D Artery Smoothed Mesh



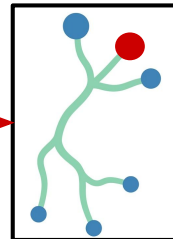
Voxel
Skeletonization



Endpoint
Detection



Ostium
Identification

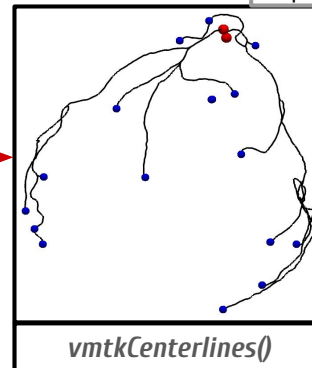


Voxel $\rightarrow \{Px, Py, Pz\}$

Artery Seed
Points



Centerline
Extraction



vmtkCenterlines()

Output Data

	.xlsx			
	Px	Py	Pz	Radius
	231.876984	225.600800	-175.701660	0.396656
	231.789917	225.614838	-175.741364	0.450688
	231.789536	225.615005	-175.741394	0.450843
	231.695126	225.656937	-175.749664	0.489475
	231.401581	225.662369	-175.718719	0.520193
	231.361008	225.662460	-175.693146	0.518145
	231.336838	225.657578	-175.675217	0.515984
	230.908340	225.769974	-175.529922	0.497722
	230.855301	225.792847	-175.534698	0.503001